# Chapter 1: Numerical Series

Calculus II

February 6, 2025

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# Section 1

# Introduction to Series

# Introduction to Series

**Problem:**
Can we add infinitely many numbers at the same time?
More precisely, given a sequence $\{a_n\}$, we can form the formal sum

$$a_1 + a_2 + a_3 + \cdots = \sum_{n=1}^{\infty} a_n$$

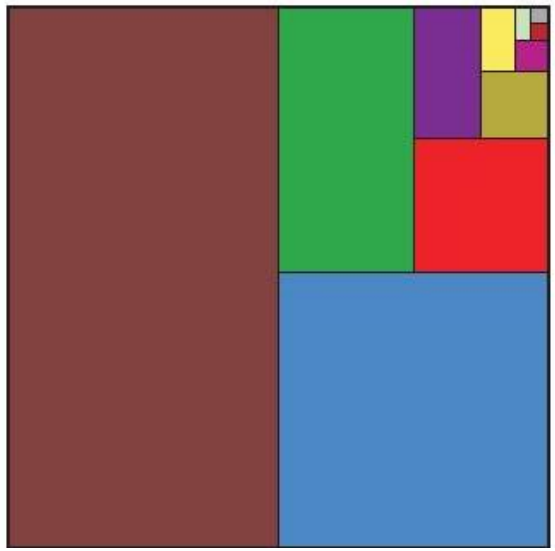
which is called a series?
**Question:**
What does this formal sum represent? Does it have a value?

# Introduction to Series

**Example:** What is

$$\begin{aligned}\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}+\frac{1}{64}+\frac{1}{128}+\cdots \\ =\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\cdots+\frac{1}{2^n}+\cdots?\end{aligned}$$

# Introduction to Series



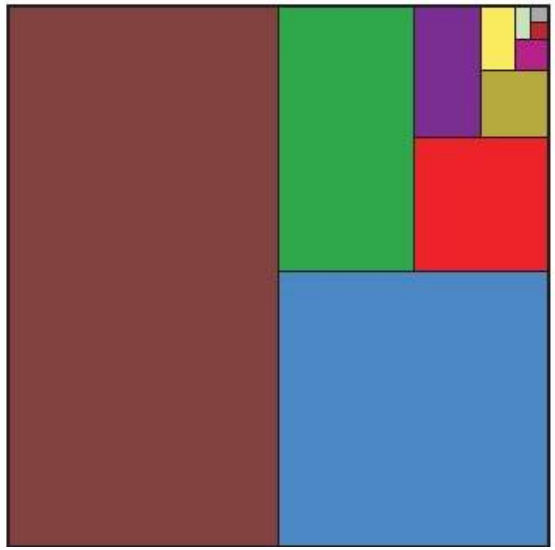
**Sum of Areas**

$$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}+\frac{1}{64}+\frac{1}{128}+\frac{1}{256}+\frac{1}{512}+\frac{1}{1024}+\cdots+\frac{1}{2^k}$$

**Total Area Covered**

$$1-\frac{1}{2^k}$$

# Introduction to Series



**Note:**

$$\lim_{k\to\infty} \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^k} = \lim_{k\to\infty} \sum_{n=1}^{k} \frac{1}{2^n}$$

$$= \lim_{k\to\infty} 1 - \frac{1}{2^k}$$

$$= 1$$

# Series

## Definition 1.1.1 (Series)

Given a sequence $\{a_n\}$, the formal sum

$$a_1 + a_2 + a_3 + a_4 + \cdots + a_n + \cdots$$

is called a series. (The series is called formal because we have not yet given it a meaning numerically.)

- The $a_n$'s are called the terms of the series.
- For each term $a_n$, the number $n$ is called the index of the term.

We denote the series by

$$\sum_{n=1}^{\infty} a_n$$

# Convergent/Divergent Series

## Definition 1.1.2 (Convergent Series)

Given a sequence $\{a_n\} = \{a_1, a_2, a_3, \ldots\}$, we define the $k$-th partial sum $S_k$ of the series $\sum_{n=1}^{\infty} a_n$ by

$$S_k = a_1 + a_2 + \cdots + a_k = \sum_{n=1}^{k} a_n$$

We say that the series $\sum_{n=1}^{\infty} a_n$ converges if the sequence of partial sums $\{S_k\}$ converges. In this case, we write

$$\sum_{n=1}^{\infty} a_n = \lim_{k \to \infty} S_k$$

Otherwise, we say that the series diverges and the sum has no defined value.

## Example 1.1.3

Suppose $a_n = \frac{1}{2^n}$. We know that

$$S_k = \sum_{n=1}^{k} \frac{1}{2^n} = 1 - \frac{1}{2^k} \to 1$$

Hence, $\sum_{n=1}^{\infty} \frac{1}{2^n}$ converges with

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = 1$$

## Example 1.1.4

Suppose $a_n = (-1)^n$. Then

$$\begin{aligned} S_1 &= 1 \\ S_2 &= 1 - 1 = 0 \\ S_3 &= 1 - 1 + 1 = 1 \\ S_4 &= 1 - 1 + 1 - 1 = 0 \end{aligned}$$

We get

$$S_k = 1 + (-1) + 1 + \cdots + (-1)^{k+1} = \begin{cases} 1 & \text{if } k \text{ is odd} \\ 0 & \text{if } k \text{ is even.} \end{cases}$$

Thus, $\{S_k\} = \{1, 0, 1, 0, 1, 0, \cdots\}$ diverges and the series

$$\sum_{n=1}^{\infty} (-1)^n \quad \text{diverges}$$

## Example 1.1.5

Determine if the series

$$\sum_{n=1}^{\infty} \frac{1}{n^2+n}$$

converges or diverges.

**Solution:** Observe that

$$a_n = \frac{1}{n^2+n} = \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

so

$$\begin{aligned} S_k &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{k} - \frac{1}{k+1}\right) \\ &= 1 - \left(\cancel{\frac{1}{2}} - \cancel{\frac{1}{2}}\right) - \left(\cancel{\frac{1}{3}} - \cancel{\frac{1}{3}}\right) - \left(\cancel{\frac{1}{4}} - \cancel{\frac{1}{4}}\right) - \cdots - \left(\cancel{\frac{1}{k}} - \cancel{\frac{1}{k}}\right) - \frac{1}{k+1} \\ &= 1 - 0 - 0 - 0 - \cdots - 0 - \frac{1}{k+1} = 1 - \frac{1}{k+1} \to 1. \end{aligned}$$

The series converges.

## Definition 1.1.6 (Geometric Series)

Let $r \in \mathbb{R}$. Then

$$\sum_{n=0}^{\infty} r^n = 1 + r + r^2 + r^3 + \cdots$$

is called a geometric series of radius $r$.

**Problem:** For which $r$ does the geometric series $\sum_{n=0}^{\infty} r^n$ converge?
To answer this question, we must look at its sequence of partial sums:

$$S_k = \sum_{n=0}^{k} r^n = 1 + r + r^2 + \cdots + r^k$$

convergent ?

$$\frac{1 - r^{k+1}}{1 - r} = S_k$$

$(r \neq 1)$

# Geometric Series

**Problem:** For which $r$ does the series $\sum_{n=0}^{\infty} r^n$ converge?

**Case 1:** $r = 1$

$$S_k = 1 + 1 + 1 + \cdots + 1 = k + 1$$

Since $\{S_k\} = \{k + 1\}$ diverges, the series $\sum_{n=0}^{\infty} 1^n$ diverges.

# Geometric Series

**Problem:** For which $r$ does the series $\sum_{n=0}^{\infty} r^n$ converge?

**Case 2:** $r = -1$

$$S_k = 1 + (-1) + 1 + \cdots + (-1)^k = \begin{cases} 1 & \text{if } k \text{ is even,} \\ 0 & \text{if } k \text{ is odd} \end{cases}$$

Since $\{S_k\} = \{1, 0, 1, 0, 1, 0, \ldots\}$ diverges, $\sum_{n=0}^{\infty} (-1)^n$ diverges.

# Geometric Series

**Problem:** For which $r$ does the series $\sum_{n=0}^{\infty} r^n$ converge?
**Case 3:** $r \neq 1$

$$\begin{aligned} S_k &= 1 + r + r^2 + \cdots + r^k \\ rS_k &= r + r^2 + \cdots + r^k + r^{k+1} \\ \Rightarrow (1-r)S_k &= 1 - r^{k+1} \\ \Rightarrow S_k &= \frac{1-r^{k+1}}{1-r} \end{aligned}$$

$\frac{1-r^{k+1}}{1-r} = S_k$.

formula.

# Geometric Series

**Problem:** For which $r$ does the series $\sum_{n=0}^{\infty} r^n$ converge?

Now

$$\left|r^{k+1}\right| \to \begin{cases} 0 & \text{if } |r| < 1 \\ \infty & \text{if } |r| > 1 \end{cases}$$

But if $S_k = \frac{1-r^{k+1}}{1-r}$,

$$\lim_{k\to\infty} S_k = \begin{cases} \frac{1}{1-r} & \text{if } |r| < 1 \\ \text{does not exist} & \text{if } |r| \geq 1 \end{cases}$$

# Geometric Series

## Theorem 1.1.7 (Geometric Series Test)

*A geometric series* $\sum_{n=0}^{\infty} r^n$ *converges if and only if* $|r| < 1$. *Moreover, if* $|r| < 1$,

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

**Example:**

$$\sum_{n=0}^{\infty} \frac{1}{2^n} = \frac{1}{1-\frac{1}{2}} = 2$$

# Divergence Test

**Problem:** Does $\sum_{n=1}^{\infty} \frac{n}{n+1}$ converge?

**Observation:** Assume that $\sum_{n=1}^{\infty} a_n$ converges to $L$ and that

$$S_k = a_1 + a_2 + \cdots + a_k = \sum_{n=1}^{k} a_n$$

Then

$$S_{k+1} - S_k = \sum_{n=1}^{k+1} a_n - \sum_{n=1}^{k} a_n = a_{k+1}$$

It follows that

$$\lim_{k\to\infty} a_k = \lim_{k\to\infty} a_{k+1} = \lim_{k\to\infty} S_{k+1} - S_k = \lim_{k\to\infty} S_{k+1} - \lim_{k\to\infty} S_k$$
$$= L - L = 0.$$

If $a_n = \frac{n}{n+1}$, then $a_n \to 1 \neq 0$, so $\sum_{n=1}^{\infty} \frac{n}{n+1}$ diverges.

# Divergence Test

## Theorem 1.1.8 (Divergence Test)

*Assume that* $\sum_{n=1}^{\infty} a_n$ *converges, then*

$$\lim_{n\to\infty} a_n = 0$$

*In particular, if* $\lim_{n\to\infty} a_n$ *does not exist or*

$$\lim_{n\to\infty} a_n \neq 0,$$

*then* $\sum_{n=1}^{\infty} a_n$ *diverges.*

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} a_{n+1}$

$= \lim_{n\to\infty}(S_{n+1} - S_n)$

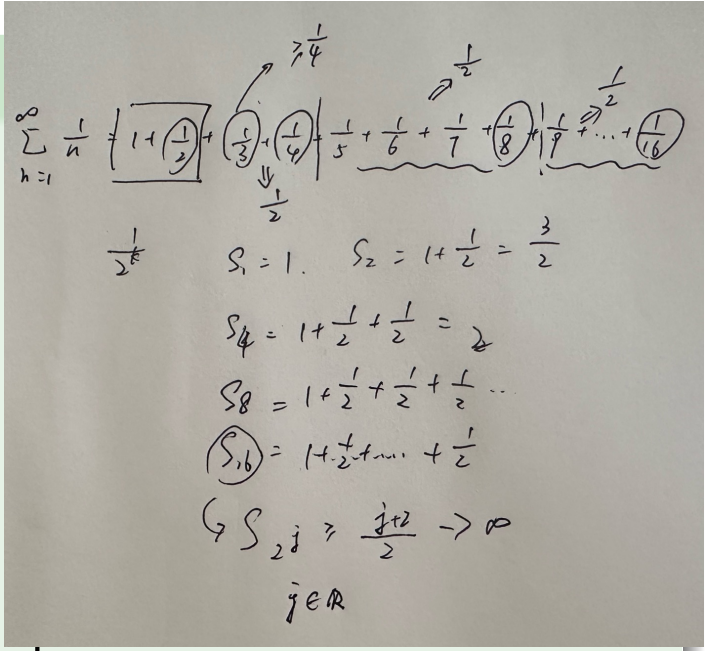
$= \lim_{n\to\infty} S_{n+1} - \lim_{n\to\infty} S_n$

$= L - L = 0.$

**Question:** Does the converse hold? That is, if $\lim_{k\to\infty} a_k = 0$, does $\sum_{n=1}^{\infty} a_n$ converge?

## Example 1.1.9

Consider $\sum_{n=1}^{\infty} \frac{1}{n}$.



$$S_1 = 1 = \frac{2}{2}$$

$$S_2 = 1 + \frac{1}{2} = \frac{3}{2}$$

$$S_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$$

$$\geq 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) = \frac{1}{2}$$

$$S_8 = S_4 + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}$$

$$\geq \frac{4}{2} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{5}{2}$$

**Observation:** For each $j \in \mathbb{N}$

$$S_{2^j} \geq \frac{j+2}{2} \to \infty$$

# Divergence Test

We have shown that for $\sum_{n=1}^{\infty} \frac{1}{n}$, the sequence $\{S_k\}$ of partial sums is unbounded and therefore divergent. Hence,

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

even though

$$\lim_{n\to\infty} \frac{1}{n} = 0$$

## Definition 1.1.10 (Harmonic Series)

The divergent series

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

is called the Harmonic Series.

# Section 2

# Arithmetic of Series

# Arithmetic of Series

**Question:** Assume that $\sum_{n=1}^{\infty} a_n$ converges, what can we say about the series $\sum_{n=1}^{\infty} 2 \cdot a_n$?

$$S_k = \sum_{n=1}^{k} a_n \quad \text{and} \quad T_k = \sum_{n=1}^{k} 2 \cdot a_n$$

then

$$T_k = \sum_{n=1}^{k} 2 \cdot a_n = 2 \cdot \sum_{n=1}^{k} a_n = 2 \cdot S_k$$

Since $\sum_{n=1}^{\infty} a_n$ converges, $\lim_{k\to\infty} S_k$ exists. Let

$$\lim_{k\to\infty} S_k = \sum_{n=1}^{\infty} a_n = S$$

Then

$$\lim_{k\to\infty} T_k = \lim_{k\to\infty} 2 \cdot S_k = 2 \cdot \lim_{k\to\infty} S_k = 2 \cdot S$$

Therefore, $\sum_{n=1}^{\infty} 2 \cdot a_n$ also converges and

$$\sum_{n=1}^{\infty} 2 \cdot a_n = 2 \cdot \sum_{n=1}^{\infty} a_n$$

# Arithmetic of Series

## Theorem 1.2.1 (Arithmetic Rules for Series I)

*Assume that* $\sum_{n=1}^{\infty} a_n$ *and* $\sum_{n=1}^{\infty} b_n$ *both converge.*

1. *The series* $\sum_{n=1}^{\infty} ca_n$ *converges for every* $c \in \mathbb{R}$ *and*

$$\sum_{n=1}^{\infty} ca_n = c \sum_{n=1}^{\infty} a_n$$

2. *The series* $\sum_{n=1}^{\infty} (a_n + b_n)$ *converges and*

$$\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n.$$

# Arithmetic of Series

**Remark:** Given a series $\sum_{n=1}^{\infty} a_n$ and $j \in \mathbb{N}$, let

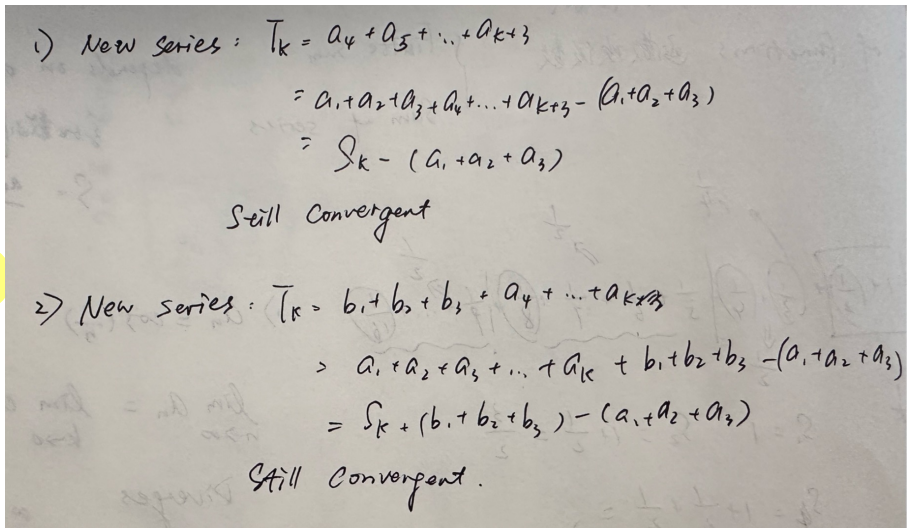
$$\sum_{n=j}^{\infty} a_n = a_j + a_{j+1} + a_{j+2} + a_{j+3} + \cdots$$

We say that $\sum_{n=j}^{\infty} a_n$ converges if

$$\lim_{k \to \infty} T_k$$

exists, where

$$T_k = \sum_{n=j}^{j+k-1} a_n = a_j + a_{j+1} + a_{j+2} + a_{j+3} + \cdots + a_{j+k-1}$$



# Arithmetic of Series

## Theorem 1.2.2 (Arithmetic Rules for Series II)

*1. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=j}^{\infty} a_n$ also converges for each $j$.*
*2. If $\sum_{n=j}^{\infty} a_n$ converges for some $j$, then $\sum_{n=1}^{\infty} a_n$ converges.*
*In either of the above two cases,*

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \cdots + a_{j-1} + \sum_{n=j}^{\infty} a_n$$

**Important Observation:** If we have a sequence

$$a_1, a_2, a_3, \cdots, a_n, \cdots$$

and we change the first $j-1$ terms to make a new sequence

$$b_1, b_2, b_3, \cdots, b_n, \cdots$$

where $b_n = a_n$ if $n \geq j$. Then the series $\sum_{n=j}^{\infty} a_n$ and $\sum_{n=j}^{\infty} b_n$ are identical. Hence **either both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge or they both diverge!!!**

# Section 3

# Positive Series

# Positive Series

## Definition 1.3.1 (Positive Series)

A series $\sum_{n=1}^{\infty} a_n$ is called positive if $a_n \geq 0$ for all $n \in \mathbb{N}$.

**Note:** If $\sum_{n=1}^{\infty} a_n$ is a positive series, then

$$\begin{aligned} S_{k+1} - S_k &= \sum_{n=1}^{k+1} a_n - \sum_{n=1}^{k} a_n \\ &= (a_1 + a_2 + \cdots + a_k + a_{k+1}) - (a_1 + a_2 + \cdots + a_k) \\ &= (a_1 + a_2 + \cdots + a_k + a_{k+1}) - (a_1 + a_2 + \cdots + a_k) \\ &= a_{k+1} \\ &\geq 0 \end{aligned}$$

Hence $\{S_k\}$ is an increasing sequence.

# Positive Series

**Key Observation:**

If $\sum_{n=1}^{\infty} a_n$ is a positive series with partial sums $S_k = \sum_{n=1}^{k} a_n$, then

$$\text{MCT} \Rightarrow \begin{cases} \{S_k\} \text{ converges} & \text{if } \{S_k\} \text{ is bounded,} \\ \{S_k\} \text{ diverges to } \infty & \text{otherwise.} \end{cases}$$

$$\Rightarrow \begin{cases} \sum_{n=1}^{\infty} a_n \text{ converges} & \text{if } \{S_k\} \text{ is bounded,} \\ \sum_{n=1}^{\infty} a_n \text{ diverges to } \infty & \text{otherwise.} \end{cases}$$

## Example 1.3.2

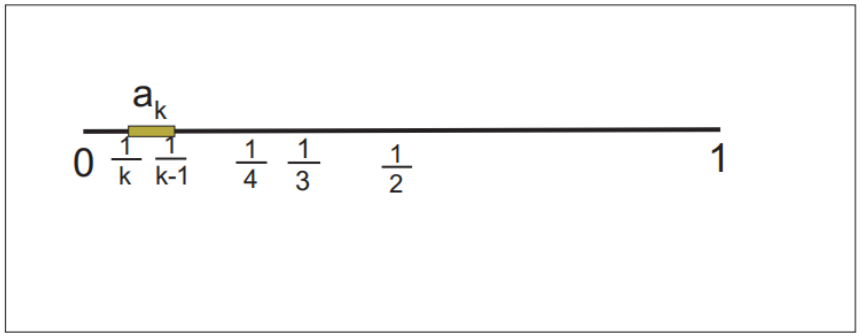
Show that $\sum_{n=2}^{\infty} \frac{1}{n^2-n}$ converges.
Note that

$$a_n = \frac{1}{n^2-n} = \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$$

so

$$\sum_{n=2}^{\infty} \frac{1}{n^2-n} = \sum_{n=2}^{\infty} \left( \frac{1}{n-1} - \frac{1}{n} \right)$$

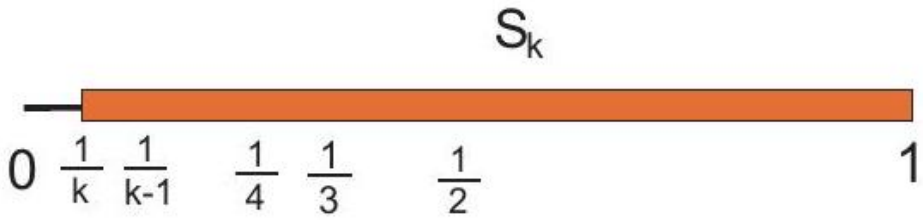
# Example




**Geometric Solution:**

$$a_2 = \frac{1}{1} - \frac{1}{2} \rightarrow \text{ distance from } 1 \text{ to } \frac{1}{2}$$
$$a_3 = \frac{1}{2} - \frac{1}{3} \rightarrow \text{ distance from } \frac{1}{2} \text{ to } \frac{1}{3}$$
$$a_4 = \frac{1}{3} - \frac{1}{4} \rightarrow \text{ distance from } \frac{1}{3} \text{ to } \frac{1}{4}.$$
$$\vdots$$
$$a_k = \frac{1}{k-1} - \frac{1}{k} \rightarrow \text{ distance from } \frac{1}{k-1} \text{ to } \frac{1}{k}.$$

# Example




$$S_2 = 1 - \frac{1}{2}$$

$$S_3 = S_2 + \left(\frac{1}{2} - \frac{1}{3}\right) = 1 - \frac{1}{3}$$

$$S_4 = S_3 + \left(\frac{1}{3} - \frac{1}{4}\right) = 1 - \frac{1}{4}$$

$$S_k = S_{k-1} + \left(\frac{1}{k-1} - \frac{1}{k}\right) = 1 - \frac{1}{k}$$

# Example

**Algebraic Solution:**

$$
\begin{aligned}
S_k &= \left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right) \\
&\quad \cdots+\left(\frac{1}{k-2}-\frac{1}{k-1}\right)+\left(\frac{1}{k-1}-\frac{1}{k}\right) \\
&= \left(1-\frac{\cancel{1}}{\cancel{2}}\right)+\left(\frac{\cancel{1}}{\cancel{2}}-\frac{\cancel{1}}{\cancel{3}}\right)+\left(\frac{\cancel{1}}{\cancel{3}}-\frac{\cancel{1}}{\cancel{4}}\right) \\
&\quad \cdots+\left(\frac{1}{k-2}-\frac{1}{k-1}\right)+\left(\frac{1}{k-1}-\frac{1}{k}\right) \\
&= 1-\frac{1}{k}
\end{aligned}
$$

# Example

**Conclusion:** The series

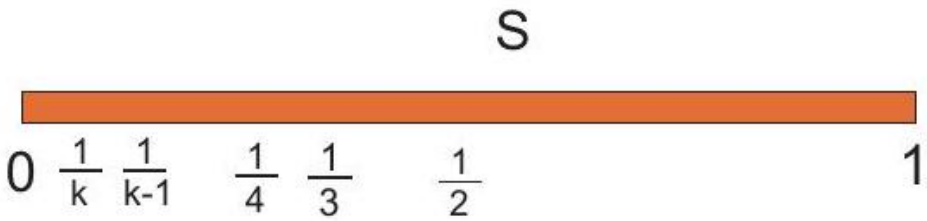
$$\sum_{n=2}^{\infty} \frac{1}{n^2 - n}$$

is positive and

$$S_k = 1 - \frac{1}{k} < 1$$

for each $k \geq 2$, hence the series converges.

# Example




We have

$$\sum_{n=2}^{\infty} \frac{1}{n^2-n} = \lim_{k\to\infty} \sum_{n=2}^{k} \frac{1}{n^2-n}$$
$$= \lim_{k\to\infty} 1 - \frac{1}{k}$$
$$= 1$$

**Problem:** Does $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converge or diverge?

positive series

**Key Observation:** We saw that

$$\sum_{n=2}^{\infty} \frac{1}{n^2 - n} = 1$$

and that if $n \geq 2$, then

$$\frac{1}{n^2} < \frac{1}{n^2 - n}$$

If

$$T_k = \sum_{n=1}^{k} \frac{1}{n^2} \text{ and } S_k = \sum_{n=2}^{k} \frac{1}{n^2 - n}$$

then for $k \geq 2$,

$$T_k = 1 + \sum_{n=2}^{k} \frac{1}{n^2} \leq 1 + \sum_{n=2}^{k} \frac{1}{n^2 - n} < 2$$

Hence $\{T_k\}$ is bounded above by 2 and $\{T_k\}$ converges, as does

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

$a_n = \frac{1}{n^2} > 0$

$\sum_{n=1}^{\infty} a_n$ $\Leftrightarrow$ $S_k$ is bounded.

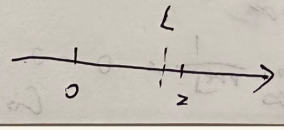
$\frac{1}{n^2} = \frac{1}{n \cdot n} \le \frac{1}{(n-1)n}$ $n \ge 2$

find a bound.

$S_k = \frac{1}{1} + \frac{1}{2^2} + \cdots + \frac{1}{k^2} \le (1 + \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{(k-1)k}$

$= 1 + 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots + \frac{1}{k-1} - \frac{1}{k} = 2 - \frac{1}{k} \le 2$

$S_k$.




$\Rightarrow$ $S_k$ is convergent.

Comparison Test

# Comparison Test

**Question:** If $0 \leq a_n \leq b_n$ and $\sum_{n=1}^{\infty} b_n$ converges, must $\sum_{n=1}^{\infty} a_n$ converge?

## Theorem 1.3.3 (The Comparison Test for Series)

*Assume that* $0 \leq a_n \leq b_n$ *for each* $n \in \mathbb{N}$.

1. *If* $\sum_{n=1}^{\infty} b_n$ *converges, then* $\sum_{n=1}^{\infty} a_n$ *converges.*
2. *If* $\sum_{n=1}^{\infty} a_n$ *diverges, then* $\sum_{n=1}^{\infty} b_n$ *diverges.*

( $b_n$ bounded, $\Rightarrow a_n$ must bounded)

# Comparison Test

**Proof of the Comparison Test:**

Assume that $0 \leq a_n \leq b_n$ for each $n \in \mathbb{N}$. Let

$$T_k = \sum_{n=1}^{k} a_n \text{ and } S_k = \sum_{n=1}^{k} b_n$$

1) Assume that $\sum_{n=1}^{\infty} b_n$ converges with $\sum_{n=1}^{\infty} b_n = S$. Then

$$T_k = \sum_{n=1}^{k} a_n \leq S_k = \sum_{n=1}^{k} b_n \leq S$$

for each $k \in \mathbb{N}$. It follows that $\{T_k\}$ is bounded and that

$$\sum_{n=1}^{\infty} a_n$$

also converges.

# Comparison Test

**Proof of the Comparison Test:**

Assume that $0 \leq a_n \leq b_n$ for each $n \in \mathbb{N}$. Let

$$T_k = \sum_{n=1}^{k} a_n \text{ and } S_k = \sum_{n=1}^{k} b_n$$

HW

2) Assume that $\sum_{n=1}^{\infty} a_n$ diverges. Let $M > 0$. Then we can find an $N \in \mathbb{N}$ so that $M \leq T_N$. But if $k \geq N$,

$$M \leq T_N \leq S_N \leq S_k$$

and hence

$$\sum_{n=1}^{\infty} b_n$$

also diverges.

# Comparison Test

**Three Important Observations:**

1) If $\sum_{n=1}^{\infty} a_n$ converges, then we cannot say anything about $\sum_{n=1}^{\infty} b_n$.
2) If $\sum_{n=1}^{\infty} b_n$ diverges, then we cannot say anything about $\sum_{n=1}^{\infty} a_n$.
3) Since the first few terms do not affect whether or not a series diverges, for the Comparison Test to hold, we really only need that

$$\mathbf{0} \leq a_n \leq b_n$$

for each $n \geq k$, where $K \in \mathbb{N}$. That is, the conditions of the theorem need only be satisfied by the elements of the tails of the two sequences.

## Example 1.3.4

We have seen that the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges and that

$$0 < \frac{1}{n^3} \leq \frac{1}{n^2}$$

for all $n \in \mathbb{N}$.
If $a_n = \frac{1}{n^3}$ and $b_n = \frac{1}{n^2}$, then the Comparison Test shows that the series

$$\sum_{n=1}^{\infty} \frac{1}{n^3}$$

also converges.
In fact, if $p \geq 2$, then

$$0 < \frac{1}{n^p} \leq \frac{1}{n^2}$$

so

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges.

Since $\frac{1}{n^2} < \frac{1}{(n-1)n}$

# Examples

**Note:** We know that $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges and that

$$0 < \frac{1}{n} \leq \frac{1}{\sqrt{n}}$$

for all $n \in \mathbb{N}$. If $a_n = \frac{1}{n}$ and $b_n = \frac{1}{\sqrt{n}}$, then the Comparison Test shows that the series

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

also diverges. In fact, if $p \leq 1$, then

$$0 < \frac{1}{n} \leq \frac{1}{n^p}$$

so

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

diverges.

# Examples

**Question:** We know that the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges if $p \leq 1$ and it converges if $p \geq 2$. What happens if

$$1 < p < 2$$

**Note:** If $1 < p < 2$, then

$$\frac{1}{n^2} \leq \frac{1}{n^p} \leq \frac{1}{n}$$

so the Comparison Test fails!

## Example 1.3.5

Show that the series

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n^2}\right)$$

converges. We know that $0 < \sin(x) < x$ for all $x \in (0, 1)$ so

$$0 < \sin\left(\frac{1}{n^2}\right) < \frac{1}{n^2}$$

for all $n \in \mathbb{N}$. If $a_n = \sin\left(\frac{1}{n^2}\right)$ and $b_n = \frac{1}{n^2}$, then the Comparison Test shows that the series

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n^2}\right)$$

also converges.

# Failure of the Comparison Test

**Question:** Does the series

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$$

converge? We know that

$$0 < \sin\left(\frac{1}{n}\right) \leq \frac{1}{n}$$

for all $n \in \mathbb{N}$ but $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges so the Comparison Test fails. Since $\lim_{n\to\infty} \frac{1}{n} = 0$, the Fundamental Trig Limits shows that

$$\lim_{n\to\infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}} = 1$$

so for large $n$ we have

$$\sin\left(\frac{1}{n}\right) \cong \frac{1}{n}$$

Does this mean that

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$$

also diverges?

# The Limit Comparison Test

## Theorem 1.3.6 (The Limit Comparison Test for Series)

*Assume that $a_n > 0$ and $b_n > 0$ for each $n \in \mathbb{N}$. Assume also that*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = L$$

finite

*where either $0 \leq L \in \mathbb{R}$ or $L = \infty$.* infinite

1. *If $0 < L < \infty$, then $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{n=1}^{\infty} b_n$ converges.*
2. *If $L = 0$ and $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.*
   *Equivalently, if $\sum_{n=1}^{\infty} a_n$ diverges, then so does $\sum_{n=1}^{\infty} b_n$.*
3. *If $L = \infty$ and $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} b_n$ converges.*
   *Equivalently, if $\sum_{n=1}^{\infty} b_n$ diverges, then so does $\sum_{n=1}^{\infty} a_n$.*

$0$ $b \to a$. convergent
$a \to b$ divergent
$\infty$ $b \to a$ divergent
$a \to b$ convergent.

# The Limit Comparison Test

**Proof of the Limit Comparison Test:** First we assume that

$$\lim_{n\to\infty} \frac{a_n}{b_n} = L$$

1) If $0 < L < \infty$, the interval $\left(\frac{L}{2}, 2L\right)$ is an open interval containing $L$. It follows that we can find a cutoff $N \in \mathbb{N}$ so that if $n \geq N$, then

$$\frac{L}{2} < \frac{a_n}{b_n} < 2L$$

or equivalently that

$$\frac{L}{2} \cdot b_n < a_n < 2Lb_n$$

If $\sum_{n=1}^{\infty} a_n$ converges, then the Comparison Test shows that

$$\sum_{n=1}^{\infty} \frac{L}{2} \cdot b_n$$

converges and hence so does

$$\sum_{n=1}^{\infty} b_n$$

# The Limit Comparison Test

**Proof of the Limit Comparison Test (continued):**

1) Since

$$\frac{L}{2} \cdot b_n < a_n < 2Lb_n, \quad n \geq N,$$

if $\sum_{n=1}^{\infty} b_n$ converges, then so does

$$\sum_{n=1}^{\infty} 2L \cdot b_n$$

By the Comparison Test

$$\sum_{n=1}^{\infty} a_n$$

also converges.

# The Limit Comparison Test

**Proof of the Limit Comparison Test (continued):**

2) If $L = 0$, then we can find a cut off $N \in \mathbb{N}$ so that if $n \geq N$, then

$$0 < \frac{a_n}{b_n} < 1$$

or equivalently that

$$0 < a_n < b_n$$

If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges as well by the Comparison Test. Equivalently, if $\sum_{n=1}^{\infty} a_n$ diverges, then so does $\sum_{n=1}^{\infty} b_n$.

# The Limit Comparison Test

**Proof of the Limit Comparison Test (continued):**

3) If $L = \infty$, then we can find a cut off $N \in \mathbb{N}$ so that if $n \geq N$, then

$$\frac{a_n}{b_n} > 1$$

or equivalently that

$$b_n < a_n$$

If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} b_n$ converges as well by the Comparison Test. Equivalently, if $\sum_{n=1}^{\infty} b_n$ diverges, then so does $\sum_{n=1}^{\infty} a_n$.

# The Limit Comparison Test

**Summary:**

1) If $\lim_{n\to\infty} \frac{a_n}{b_n} = L$ where $0 < L < \infty$, then for large $n$ we have

$$\frac{a_n}{b_n} \cong L$$

or

$$a_n \cong L b_n$$

When $\lim_{n\to\infty} \frac{a_n}{b_n} = L$ where $0 < L < \infty$, we say that $a_n$ and $b_n$ have the same order of magnitude. We write

$$a_n \asymp b_n.$$

The Limit Comparison Test says that two positive series with terms of the same order of magnitude will have the same convergence properties.

# The Limit Comparison Test

**Summary:**

2) If $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$, then $b_n$ must eventually be much larger than $a_n$. In this case, we write

$$a_n \ll b_n$$

and we say that the order of magnitude of $a_n$ is smaller than the order of magnitude of $b_n$.

If the smaller series $\sum_{n=1}^{\infty} a_n$ diverges to $\infty$, it would make sense that $\sum_{n=1}^{\infty} b_n$ also diverges to $\infty$.

3) If $\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$, then $a_n$ must eventually be much larger than $b_n$. That is

$$b_n \ll a_n.$$

This time, if the larger series $\sum_{n=1}^{\infty} a_n$ converges, it would make sense that $\sum_{n=1}^{\infty} b_n$ would converge as well.

## Example 1.3.7

The series

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$$

diverges. Since

$$\lim_{n\to\infty} \frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}} = 1$$

and since

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

diverges, the Limit Comparison Test shows that

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$$

also diverges.

## Example 1.3.8

The series $\sum_{n=1}^{\infty} \frac{2n}{n^3-n+1}$ converges.

Let $a_n = \frac{2n}{n^3-n+1}$ and $b_n = \frac{1}{n^2}$. Then

$$\frac{a_n}{b_n} = \frac{\frac{2n}{n^3-n+1}}{\frac{1}{n^2}}$$

$$= \frac{2n^3}{n^3-n+1}$$

$$= \frac{n^3}{n^3}\left(\frac{2}{1-\frac{1}{n^2}+\frac{1}{n^3}}\right)$$

$$= \frac{2}{1-\frac{1}{n^2}+\frac{1}{n^3}}$$

Therefore,

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{2}{1-\frac{1}{n^2}+\frac{1}{n^3}} = \frac{2}{1} = 2$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, the Limit Comparison Test shows that $\sum_{n=1}^{\infty} \frac{2n}{n^3-n+1}$ converges.

# $p$-Series Revisited

**Recall:** We know that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges while $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges. We also know that if $1 < p < 2$, then

$$\frac{1}{n^2} \leq \frac{1}{n^p} \leq \frac{1}{n}$$

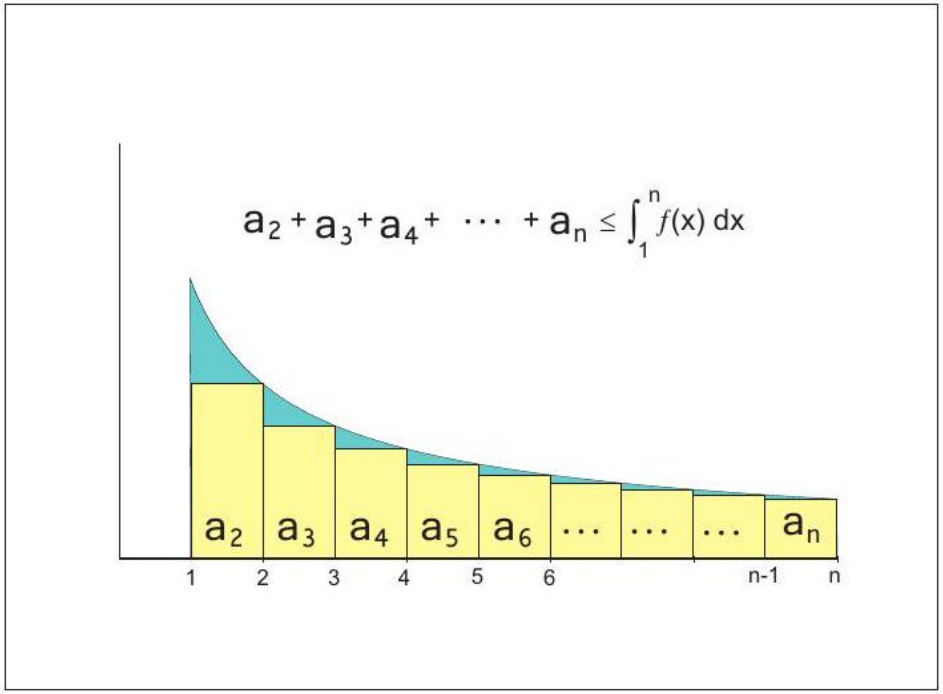
so the Comparison Test (and Limit Comparison Test) fail to determine if

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges.

**Key Idea:** There is a close relationship between series and improper integrals.

# Integral Test



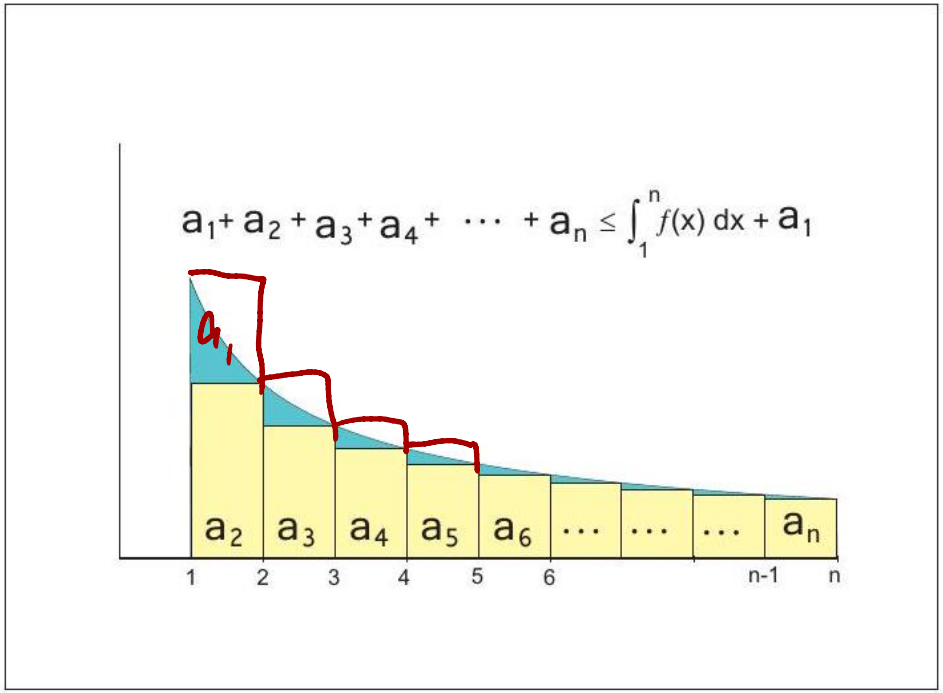

Assume that $f$ is continuous, decreasing and $f(x) > 0$ on $[1, \infty)$. For each $k \in \mathbb{N}$, let

$$a_k = f(k) \quad \text{and} \quad S_n = \sum_{k=1}^{n} a_k$$

We claim

$$\int_1^{n+1} f(x)dx \leq S_n \leq a_1 + \int_1^n f(x)dx.$$

# Integral Test




$a_1 \le \int_1^2 f(x)dx \le a_1 \le a_1$

$\int_n^{n-1} f(x)dx$

$n=1,\ n-1=0$

$\Rightarrow \int_1^1 f(x)dx = a_1$

Assume that $f$ is continuous, decreasing and $f(x) > 0$ on $[1, \infty)$. For each $k \in \mathbb{N}$, let

$$a_k = f(k) \quad \text{and} \quad S_n = \sum_{k=1}^{n} a_k$$

We claim

$$\int_n^{n+1} f(x)dx \le a_n \le \int_{n-1}^{n} f(x)dx$$

$(\Rightarrow)$

$$\int_1^{n+1} f(x)dx \le S_n \le a_1 + \int_1^n f(x)dx.$$

## Theorem 1.3.9 (The Integral Test)

*Assume that $f$ is continuous on $[1,\infty)$, $f(x) > 0$ on $[1,\infty)$, and $f$ is decreasing on $[1,\infty)$. Let*

$$a_k := f(k).$$

*1) If $S_n = \sum_{k=1}^{n} a_k$, then for all $n \in \mathbb{N}$,*

$$\int_1^{n+1} f(x)dx \leq S_n \leq a_1 + \int_1^{n} f(x)dx.$$

*2) $\sum_{k=1}^{\infty} a_k$ converges if and only if $\int_1^{\infty} f(x)dx$ converges.*
*3) If $\sum_{k=1}^{\infty} a_k$ converges, with $S = \sum_{k=1}^{\infty} a_k$, then*

$$\int_1^{\infty} f(x)dx \leq \sum_{k=1}^{\infty} a_k \leq a_1 + \int_1^{\infty} f(x)dx$$

*and*

$$\int_{n+1}^{\infty} f(x)dx \leq S - S_n \leq \int_n^{\infty} f(x)dx$$

**Note:**

1) We have already established that

$$\int_1^{n+1} f(x)dx \leq S_n \leq a_1 + \int_1^n f(x)dx$$

2) If $\sum_{k=1}^{\infty} a_k$ converges, then

$$\int_1^{n+1} f(x)dx \leq \sum_{k=1}^{\infty} a_k$$

for all $\boldsymbol{n} \in \mathbb{N}$, so

$$\int_1^{\infty} f(x)dx$$

converges by the Monotone Convergence Theorem. If $\int_1^{\infty} f(x)dx$ converges, then

$$\sum_{k=1}^{n} a_k \leq a_1 + \int_1^{\infty} f(x)dx$$

for all $\boldsymbol{n} \in \mathbb{N}$, so

$$\sum_{k=1}^{\infty} a_k$$

converges by the Monotone Convergence Theorem.

# Integral Test

**Note:**

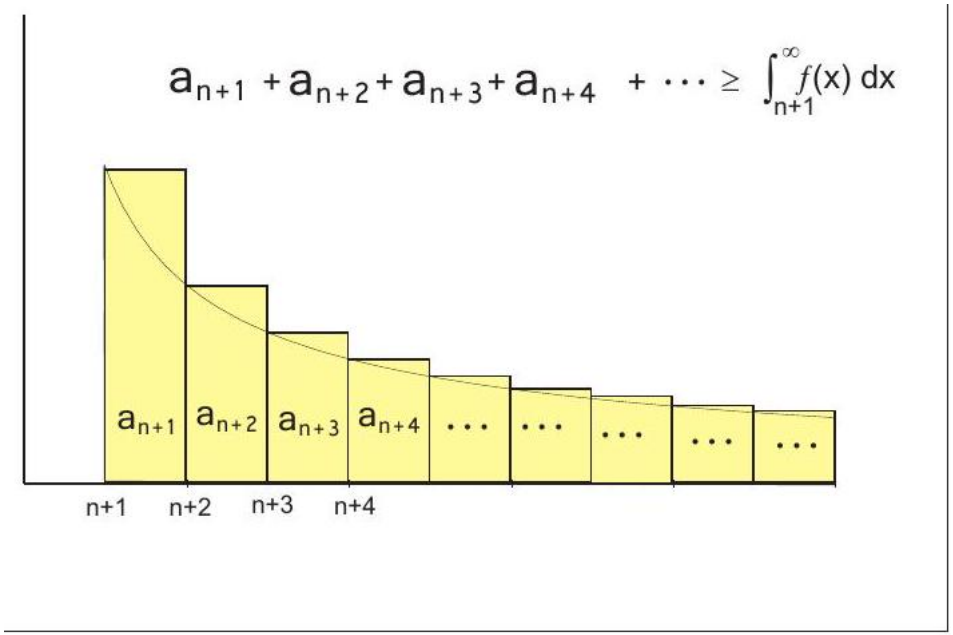
3) If $\sum_{k=1}^{\infty} a_k$ converges, with $S = \sum_{k=1}^{\infty} a_k$, then

$$\lim_{n\to\infty} \int_1^{n+1} f(x)dx \leq \lim_{n\to\infty} \sum_{k=1}^{n} a_k \leq a_1 + \lim_{n\to\infty} \int_1^{n} f(x)dx$$

Hence

$$\int_1^{\infty} f(x)dx \leq \sum_{k=1}^{\infty} a_k \leq a_1 + \int_1^{\infty} f(x)dx$$

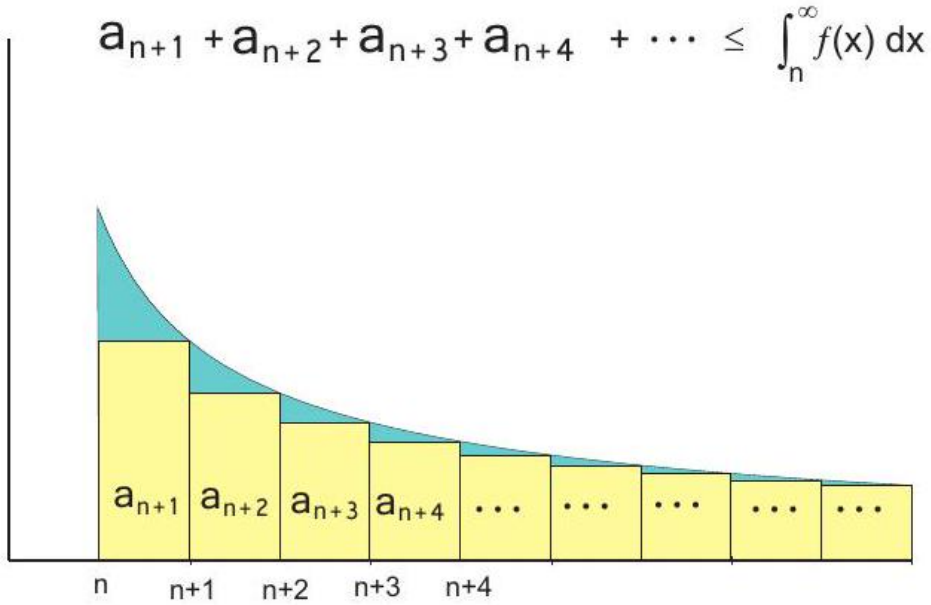
# Integral Test




Finally

$$\int_{n+1}^{\infty} f(x)dx \leq S - S_n \leq \int_{n}^{\infty} f(x)dx$$

# Integral Test




Finally

$$\int_{n+1}^{\infty} f(x)dx \leq S - S_n \leq \int_n^{\infty} f(x)dx$$

# $p$-Series Test

**Key Observation :** The function $f(x) = \frac{1}{x^p}$ satisfies the conditions of the Integral Test for $p > 0$ and

$$\int_1^\infty \frac{1}{x^p} dx$$

converges if and only if $p > 1$.

## Theorem 1.3.10 ($p$-Series Test)

*The series*

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

*converges if and only if* $p > 1$.

## Example 1.3.11

Determine whether the series

$$\sum_{n=2}^{\infty} \frac{1}{n \ln(n)}$$

converges.

**Observation:** Let $f(x) = \frac{1}{x \ln(x)}$ and let $a_n = f(n) = \frac{1}{n \ln(n)}$. Then $f$ is continuous, positive, and decreasing on $[2, \infty)$.

Hence

$$\sum_{n=2}^{\infty} a_n$$

converges if and only if

$$\int_2^{\infty} f(x)dx$$

converges.

convergent or divergent.

$a_n \quad \sum_{n=k}^{\infty} a_n \iff \int_k^{\infty} f(x)dx$ ?

## Example 1.3.12 (continued)

$$\int_2^\infty \frac{1}{x\ln(x)}dx = \lim_{b\to\infty}\int_2^b \frac{1}{x\ln(x)}dx$$

Make a change of variable $u = \ln(x)$, and so $du = \frac{1}{x}dx$ and

$$\begin{aligned}\lim_{b\to\infty}\int_2^b \frac{1}{x\ln(x)}dx &= \lim_{b\to\infty}\int_{\ln(2)}^{\ln(b)} \frac{1}{u}du \\ &= \lim_{b\to\infty} \ln(u)\Big|_{\ln(2)}^{\ln(b)} \\ &= \lim_{b\to\infty} \ln(\ln(b)) - \ln(\ln(2))\end{aligned}$$

which diverges to $\infty$. Therefore

$$\sum_{n=2}^\infty \frac{1}{n\ln(n)}$$

diverges by the Integral Test.

## Example 1.3.13

Show that $\sum_{n=2}^{\infty} \frac{1}{n(\ln(n))^2}$ converges.

**Observation:** Since $f(x) = \frac{1}{x(\ln(x))^2}$ is continuous and decreasing with $f(x) > 0$ on $[2, \infty)$, the Integral Test can be used to conclude that

$$\sum_{n=2}^{\infty} \frac{1}{n(\ln(n))^2}$$

converges if and only if

$$\int_2^{\infty} \frac{1}{x(\ln(x))^2} dx$$

converges.

## Example 1.3.14 (continued)

To evaluate $\int_2^b \frac{1}{x(\ln(x))^2}dx$, use the substitution $u = \ln(x), du = \frac{dx}{x}$ to get

$$\begin{aligned}
\int_2^\infty \frac{1}{x(\ln(x))^2}dx &= \lim_{b\to\infty}\int_2^b \frac{1}{x(\ln(x))^2}dx \\
&= \lim_{b\to\infty}\int_{\ln(2)}^{\ln(b)} \frac{1}{u^2}du \\
&= \lim_{b\to\infty} -\frac{1}{u}\Big|_{\ln(2)}^{\ln(b)} \\
&= \lim_{b\to\infty} \frac{1}{\ln(2)} - \frac{1}{\ln(b)} \\
&= \frac{1}{\ln(2)}
\end{aligned}$$

Since $\int_2^\infty \frac{1}{x(\ln(x))^2}dx$ converges, so does $\sum_{n=2}^\infty \frac{1}{n(\ln(n))^2}$.

# Section 4

# Alternating Series

# Alternating Series

**Recall:** The Harmonic Series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \cdots$$

diverges.

**Question:** What can we say about the series

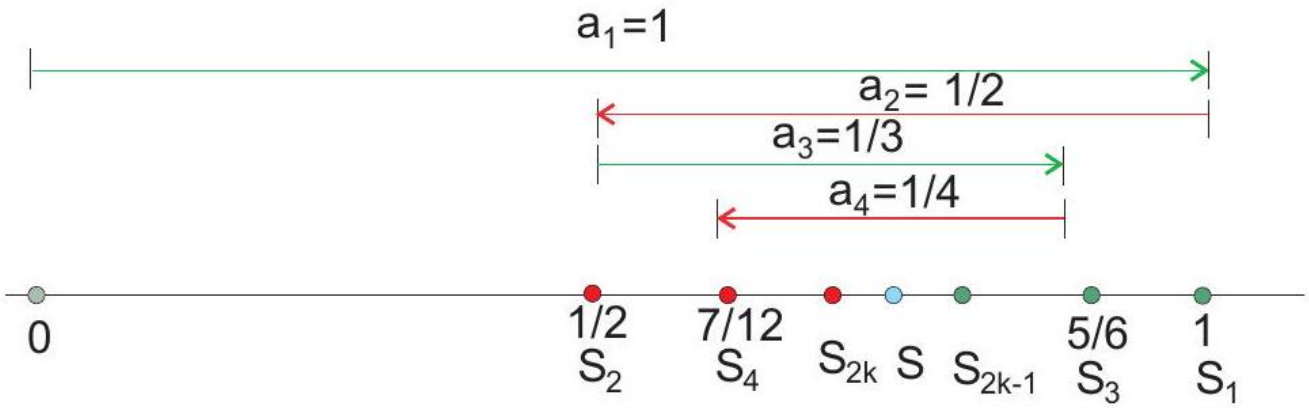
$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \cdots + (-1)^{n+1} \frac{1}{n} + \cdots ?$$

This series is one example of the Alternating Series.

Odd

Even.

# Alternating Series




Let $a_n = \frac{1}{n}$ and $S_k = \sum_{n=1}^{k} (-1)^{n+1} \frac{1}{n}$.
1) $\{S_{2k-1}\}$ is decreasing and bounded below.
2) $\{S_{2k}\}$ is increasing and bounded above.

# Alternating Series

**Claim 1:** $\{S_{2k-1}\}$ is decreasing and bounded below by 0.

- $S_{2k+1} - S_{2k-1} = \sum_{n=1}^{2k+1}(-1)^{n+1}\frac{1}{n} - \sum_{n=1}^{2k-1}(-1)^{n+1}\frac{1}{n} = -\frac{1}{2k} + \frac{1}{2k+1} < 0$ so $\{S_{2k-1}\}$ is decreasing.
- $S_{2k-1} = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{2k-3} - \frac{1}{2k-2}\right) + \frac{1}{2k-1} > 0.$

**Conclusion:** The MCT shows that $\{S_{2k-1}\}$ converges.

# Alternating Series

**Claim 2:** $\{S_{2k}\}$ is increasing and bounded above by 1.

- $S_{2k+2} - S_{2k} = \sum_{n=1}^{2k+2}(-1)^{n+1}\frac{1}{n} - \sum_{n=1}^{2k}(-1)^{n+1}\frac{1}{n} =$

$$\frac{1}{2k+1} - \frac{1}{2k+2} > 0 \text{ so } \{S_{2k}\} \text{ is increasing.}$$

- $S_{2k} = 1 + \left(-\frac{1}{2} + \frac{1}{3}\right) + \left(-\frac{1}{4} + \frac{1}{5}\right) + \cdots + \left(-\frac{1}{2k-2} + \frac{1}{2k-1}\right) - \frac{1}{2k} < 1.$

**Conclusion:** The MCT shows that $\{S_{2k}\}$ converges.

**Note**

$$|S_{2k} - S_{2k-1}| = \frac{1}{2k} \to 0$$

Therefore,

$$\lim_{k\to\infty} S_{2k-1} = S = \lim_{k\to\infty} S_{2k}$$

and $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n}$ converges (to $\ln(2)$).

# Alternating Series

## Theorem 1.4.1 (Alternate Series Test)

*Suppose that*

1) $a_n > 0$ *for all* $n \in \mathbb{N}$,
2) $a_{n+1} < a_n$ *for all* $n \in \mathbb{N}$,
3) $\lim_{n\to\infty} a_n = 0$.

*Then the series*

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n$$

*converges.*

*Moreover, if* $S_k = \sum_{n=1}^{k} (-1)^{n+1} a_n$ *and* $S = \sum_{n=1}^{\infty} (-1)^{n+1} a_n$, *then*

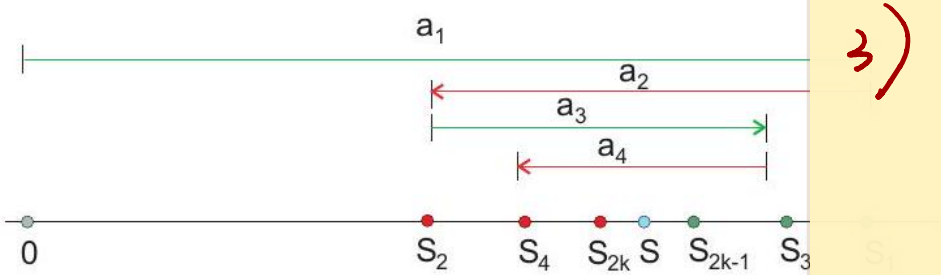
$$|S_k - S| < a_{k+1}.$$

# Alternating Series

1) $a_n > 0$

2) $a_n \downarrow$

3) $a_n \to 0$




We get

1) $\{S_{2k-1}\}$ is decreasing and bounded below by $0 \Rightarrow \{S_{2k-1}\} \to L$.

2) $\{S_{2k}\}$ is increasing and bounded above by $a_1 \Rightarrow \{S_{2k}\} \to M$.

3) $\lim_{k\to\infty} |S_{2k-1} - S_{2k}| = \lim_{k\to\infty} a_{2k} = 0 \Rightarrow L = M = S$.

Finally, since $S$ is between $S_k$ and $S_{k+1}$ for each $k$,

$$|S_k - S| < |S_k - S_{k+1}| = a_{k+1}$$

## Example 1.4.2

Show that $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n^5}$ converges to some $S$ and that

$$\left|\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n^5}-\frac{31}{32}\right|<\frac{1}{100}.$$

Solution: With $a_n=\frac{1}{n^5}$ it is clear that

1) $a_n>0$ for all $n\in\mathbb{N}$,
2) $a_{n+1}<a_n$ for all $n\in\mathbb{N}$,
3) $\lim_{n\to\infty}a_n=0$.

so the series converges by the AST.

We know that $\frac{31}{32}=1-\frac{1}{2^5}=\sum_{n=1}^{2}(-1)^{n+1}\frac{1}{n^5}=S_2$. Hence

$$\left|\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n^5}-\frac{31}{32}\right|=|S-S_2|<a_3=\frac{1}{3^5}=\frac{1}{243}<\frac{1}{100}$$

## Example 1.4.3

The Integral Test and the AST show that both

$$\sum_{n=2}^{\infty} \frac{1}{n(\ln(n))^2} \quad \text{and} \quad \sum_{n=2}^{\infty} (-1)^n \frac{1}{n(\ln(n))^2}$$

converge.
Let

$$T_k = \sum_{n=2}^{k} \frac{1}{n(\ln(n))^2} \quad \text{and} \quad T = \sum_{n=2}^{\infty} \frac{1}{n(\ln(n))^2}$$

and let

$$S_k = \sum_{n=2}^{k} (-1)^n \frac{1}{n(\ln(n))^2} \quad \text{and} \quad S = \sum_{n=2}^{\infty} (-1)^n \frac{1}{n(\ln(n))^2}$$

How large must $k$ be so that

$$|T - T_k| < \frac{1}{100} \text{ and } |S - S_k| < \frac{1}{100}$$

respectively?

# Example

**Solution:** The Integral Test shows that

$$|T - T_k| < \int_k^{\infty} \frac{1}{x(\ln(x))^2} dx = \frac{1}{\ln(k)}$$

We want

$$\frac{1}{\ln(k)} < \frac{1}{100} \Rightarrow k > e^{100} \cong 10^{43}$$

The AST shows that

$$|S - S_k| < \frac{1}{(k+1)(\ln(k+1))^2}$$

We want

$$\frac{1}{(k+1)(\ln(k+1))^2} < \frac{1}{100} \Rightarrow k \geq 14$$

# Section 4

# Alternating Series

# Absolute Convergence

**Important Observation:**

1) The Harmonic Series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, while the Alternating Series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n}$ converges even though the terms have the same order of magnitude,

$$\left|\frac{1}{n}\right| = \frac{1}{n} = \left|(-1)^{n-1}\frac{1}{n}\right|$$

2) On the other hand, both $\sum_{n=1}^{\infty} \frac{1}{n^2}$ and $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n^2}$ converge because $\frac{1}{n^2}$ is small enough!

**Question 1:** Is there a way to detect if a series will converge because its terms are small enough?

# Absolute Convergence

**Question 2:** Aside from the AST all of our tests apply to positive series. Is there any additional method for determining if an arbitrary series $\sum_{n=1}^{\infty} a_n$ converges?

## Definition 1.4.4 (Absolute Convergence)

We say that a series $\sum_{n=1}^{\infty} a_n$ converges absolutely or is absolutely convergent if the series

$$\sum_{n=1}^{\infty} |a_n| \quad \Longrightarrow \quad \sum_{n=1}^{\infty} a_n$$

converges.

**Question 3:** If $\sum_{n=1}^{\infty} a_n$ converges absolutely, does it converge?

# Absolute Convergence

## Theorem 1.4.5 (Absolute Convergence Theorem)

*Assume that the series $\sum_{n=1}^{\infty} a_n$ converges absolutely. Then $\sum_{n=1}^{\infty} a_n$ converges.*

**Proof:** Assume that

$$\sum_{n=1}^{\infty} |a_n|$$

converges. Then so does $\sum_{n=1}^{\infty} 2\,|a_n|$. Let

$$b_n = a_n + |a_n| \Rightarrow 0 \leq b_n \leq 2\,|a_n|$$

By the Comparison Test $\sum_{n=1}^{\infty} b_n$ converges.
Since $a_n = b_n - |a_n|$, it follows that $\sum_{n=1}^{\infty} a_n$ also converges.

## Example 1.4.6

Let $\{a_n\} = \left\{1, \frac{1}{2^2}, \frac{-1}{3^2}, \frac{1}{4^2}, \frac{1}{5^2}, \frac{-1}{6^2}, \cdots\right\}$. Then

$$|a_n| = \frac{1}{n^2}$$

so $\sum_{n=1}^{\infty} |a_n|$ converges and the series $\sum_{n=1}^{\infty} a_n$ converges absolutely.

## Example 1.4.7

Let $\{b_n\} = \left\{(-1)^{n+1} \sin\left(\frac{1}{\sqrt{n}}\right)\right\}$.

Since

1) $\sin\left(\frac{1}{\sqrt{n}}\right) > 0$ for all $n \in \mathbb{N}$,

2) $\sin\left(\frac{1}{\sqrt{n+1}}\right) < \sin\left(\frac{1}{\sqrt{n}}\right)$ for all $n \in \mathbb{N}$,

3) $\lim_{n\to\infty} \sin\left(\frac{1}{\sqrt{n}}\right) = 0$,

the series $\sum_{n=1}^{\infty}(-1)^{n+1} \sin\left(\frac{1}{\sqrt{n}}\right)$ converges by the AST.

However, since

$$\lim_{n\to\infty} \frac{\sin\left(\frac{1}{\sqrt{n}}\right)}{\frac{1}{\sqrt{n}}} = 1$$

and $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ diverges, $\sum_{n=1}^{\infty} \left|(-1)^{n+1} \sin\left(\frac{1}{\sqrt{n}}\right)\right| = \sum_{n=1}^{\infty} \sin\left(\frac{1}{\sqrt{n}}\right)$ diverges.

Hence $\sum_{n=1}^{\infty}(-1)^{n+1} \sin\left(\frac{1}{\sqrt{n}}\right)$ is not absolutely convergent.

# Conditional Converge

## Definition 1.4.8 (Conditional Convergence)

A series $\sum_{n=1}^{\infty} a_n$ is said to be conditionally convergent if it converges, but it is not absolutely convergent.

**Question 4:** Why do we care if a series converges absolutely rather than conditionally?

If $\sum_{n=1}^{\infty} a_n$ converges. but $\sum_{n=1}^{\infty} |a_n|$ diverges

$\Rightarrow$ conditional Converge

# Rearrangement of Series 性质 of Conditional Converge

**Question 5:** For a finite sum order does not matter. That is

$$a + b + c + d = d + c + b + a$$

Is this true for infinite sums?

## Definition 1.4.9 (Rearrangement of a Series)

Given a sequence $\{a_n\}$ and a $1-1$ and onto function $\phi : \mathbb{N} \to \mathbb{N}$, if we let

$$b_n = a_{\phi(n)}$$

then the series $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_{\phi(n)}$ is called a rearrangement of the series $\sum_{n=1}^{\infty} a_n$.

# Rearrangement of Series

**Question 6:** What do we know about the convergence of $\sum_{n=1}^{\infty} a_{\phi(n)}$ relative to that of $\sum_{n=1}^{\infty} a_n$? In particular, if $\sum_{n=1}^{\infty} a_n$ converges, does $\sum_{n=1}^{\infty} a_{\phi(n)}$ also converge with

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_{\phi(n)}?$$

# Rearrangement of Series

**Facts:**
1) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_{\phi(n)}$ for every rearrangement $\sum_{n=1}^{\infty} a_{\phi(n)}$ of $\sum_{n=1}^{\infty} a_n$.
2) If $\sum_{n=1}^{\infty} a_n$ converges conditionally, then $\sum_{n=1}^{\infty} a_{\phi(n)}$ may or may not converge.
3) If $\sum_{n=1}^{\infty} a_n$ converges conditionally and if $\alpha \in \mathbb{R} \cup \{-\infty, \infty\}$, then there exists a $1-1$ and onto function $\phi : \mathbb{N} \to \mathbb{N}$ such that

$$\sum_{n=1}^{\infty} a_{\phi(n)} = \alpha$$

**Summary:** Absolutely convergent series are stable, conditionally convergent series are not!

# Section 5

# The Ratio Test and Root Test

# Geometric Series

**Recall:**

## Geometric Series Test

A geometric series $\sum_{n=0}^{\infty} r^n$ converges if and only if $|r| < 1$.
Moreover, if $|r| < 1$,

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

## Example 1.5.1

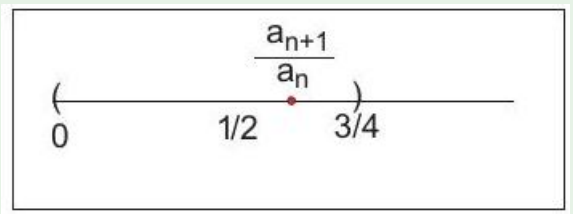
Does the series $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converge or diverge?
We know $\sum_{n=1}^{\infty} \frac{1}{2^n}$ converges, but $\frac{1}{2^n} \leq \frac{n}{2^n}$ so the Comparison Test fails even though $n \ll 2^n$.

**Key Observation:** If $a_n = \frac{n}{2^n}$, then

$$\begin{aligned}\lim_{n\to\infty} \frac{a_{n+1}}{a_n} &= \lim_{n\to\infty} \frac{\frac{n+1}{2^{n+1}}}{\frac{n}{2^n}} \\ &= \lim_{n\to\infty} \frac{n+1}{2n} \\ &= \frac{1}{2}\end{aligned}$$

We can find an $N_0$ such that if $n \geq N_0$, then

$$\frac{a_{n+1}}{a_n} < \frac{3}{4} \Rightarrow a_{n+1} < \frac{3}{4} a_n$$



# Example Cont'd

Therefore,

$$
\begin{aligned}
a_{N_0+1} &< \frac{3}{4} a_{N_0} \\
a_{N_0+2} &< \frac{3}{4} a_{N_0+1} < \left(\frac{3}{4}\right)^2 a_{N_0} \\
a_{N_0+3} &< \frac{3}{4} a_{N_0+2} < \left(\frac{3}{4}\right)^3 a_{N_0} \\
&\vdots \\
a_{N_0+k} &< \frac{3}{4} a_{N_0+(k-1)} < \left(\frac{3}{4}\right)^k a_{N_0}
\end{aligned}
$$

So

$$
\sum_{n=N_0}^{N_0+k} a_n \leq \sum_{j=0}^{k} \left(\frac{3}{4}\right)^j a_{N_0} < \frac{a_{N_0}}{1-\frac{3}{4}} \Rightarrow \sum_{n=N_0}^{\infty} a_n < \infty
$$

Therefore, $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

# Ratio Test

Basic => Resource

Observation:

Geometric Series:

$\frac{a_{n+1}}{a_n} = r \begin{cases} \in (0,1) & \checkmark \\ \in [1,\infty) & \times \end{cases}$

## Theorem 1.5.2 (Ratio Test)

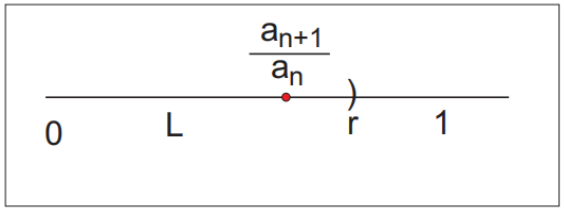
*Given a series* $\sum_{n=0}^{\infty} a_n$, *assume that*

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = L$$

*where* $0 \leq L \in \mathbb{R}$ *or* $L = \infty$.

1. *If* $0 \leq L < 1$, *then* $\sum_{n=0}^{\infty} a_n$ *converges absolutely.*
2. *If* $L > 1$, *then* $\sum_{n=0}^{\infty} a_n$ *diverges.*
3. *If* $L = 1$, *then no conclusion is possible.*

# Ratio Test




**Proof of 1):** Assume that

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L < 1$$

Let $L < r < 1$. Then we can find an $N_0$ such that if $n \geq N_0$, then

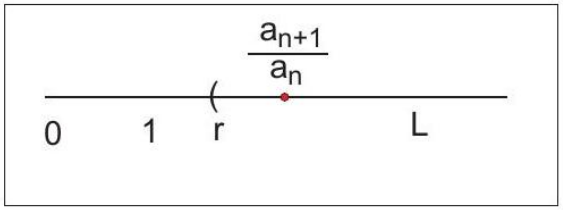
$$0 < \frac{|a_{n+1}|}{|a_n|} < r$$

We again get that for each $k \in \mathbb{N}$

$$|a_{N_0+k}| \leq |a_{N_0}|\, r^k$$

Since $0 < r < 1$, the Geometric Series Test shows that $\sum_{k=0}^{\infty} |a_{N_0}|\, r^k$ converges. The Comparison Test shows that $\sum_{n=N_0}^{\infty} |a_n|$ also converges. Hence, $\sum_{n=1}^{\infty} |a_n|$ converges.

# Ratio Test




**Proof of 2):** Assume that

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L > 1$$

Let $1 < r < L$. Then we can find an $N_0$ such that if $n \geq N_0$, then

$$\left|\frac{a_{n+1}}{a_n}\right| > r$$

If $n \geq N_0$, then

$$|a_{n+1}| \geq r \cdot |a_n| > |a_n|$$

In fact,

$$|a_{N_0+k}| \geq |a_{N_0}|\, r^k \to \infty$$

The Divergence Test shows that $\sum_{n=1}^{\infty} a_n$ diverges.

# Ratio Test

## Example 1.5.3

Let $a_n = \frac{1}{n}$ and $b_n = \frac{1}{n^2}$. Then

1) $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{\frac{1}{n+1}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n}{n+1} = 1$, and $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

2) $\lim_{n\to\infty} \frac{b_{n+1}}{b_n} = \lim_{n\to\infty} \frac{\frac{1}{(n+1)^2}}{\frac{1}{n^2}} = \lim_{n\to\infty} \frac{n^2}{n^2+2n+1} = 1$, and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

# Ratio Test

**Important notes :**

1) The Ratio Test only detects convergence if $a_n \to 0$ very rapidly.
2) The Ratio Test only detects divergence if $|a_n| \to \infty$.
3) Despite 1) and 2), the Ratio Test is probably the most important test for convergence of series.

# Ratio Test

**Question:** Does the series $\sum_{n=0}^{\infty} \frac{1}{n!}$ converge or diverge?
**Solution:** Let $a_n = \frac{1}{n!}$. Then

$$\begin{aligned}
\lim_{n\to\infty} \frac{a_{n+1}}{a_n} &= \lim_{n\to\infty} \frac{\frac{1}{(n+1)!}}{\frac{1}{n!}} \\
&= \lim_{n\to\infty} \frac{n!}{(n+1)!} \\
&= \lim_{n\to\infty} \frac{1}{n+1} \\
&= 0
\end{aligned}$$

$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{\frac{1}{(n+1)!}}{\frac{1}{n!}}\right| = \lim_{n\to\infty} \left(\frac{1}{n+1}\right) = 0 \in (0,\infty)$

converges.

The Ratio Test shows that $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges.

# Ratio Test

**Question:** Does the series $\sum_{n=0}^{\infty} \frac{1000000^n}{n!}$ converge or diverge?

**Note:** If we let

$$a_n = \frac{1000000^n}{n!}$$

then

$$a_{10} > 10^{50}$$

However,

$$\frac{a_{n+1}}{a_n} = \frac{\frac{1000000^{n+1}}{(n+1)!}}{\frac{1000000^n}{n!}} = \frac{1000000}{n+1}$$

We again have that

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{1000000}{n+1} = 0$$

so the Ratio Test show that $\sum_{n=0}^{\infty} \frac{1000000^n}{n!}$ converges.

# Ratio Test

**Question:** For which $x \in \mathbb{R}$ does the series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converge or diverge?

**Solution:** Let $a_n = \frac{x^n}{n!}$. Then if $x \neq 0$.

$$\begin{aligned}
\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} &= \lim_{n\to\infty} \frac{\frac{|x|^{n+1}}{(n+1)!}}{\frac{|x|^n}{n!}} \\
&= \lim_{n\to\infty} \frac{|x|^{n+1} n!}{|x|^n (n+1)!} \\
&= \lim_{n\to\infty} \frac{|x|}{n+1} \\
&= 0.
\end{aligned}$$

$\lim_{n\to\infty}$

$\frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n}$

$\frac{x}{n+1} = 0$

for all $x \in \mathbb{R}$

Converges

The Ratio Test shows that $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converges for all $x \in \mathbb{R}$.

**Important Observation:** We have $\lim_{n\to\infty} \frac{x^n}{n!} = 0$, so factorials eventually grow much quicker than exponentials.

# Ratio Test

**Question:** Does the series $\sum_{n=1}^{\infty} \frac{n^n}{n!}$ converge or diverge?

**Solution:** Let $a_n = \frac{n^n}{n!}$. Then

$$\begin{aligned}
\lim_{n\to\infty} \frac{a_{n+1}}{a_n} &= \lim_{n\to\infty} \frac{\frac{(n+1)^{n+1}}{(n+1)!}}{\frac{n^n}{n!}} \\
&= \lim_{n\to\infty} \frac{(n+1)^{n+1} n!}{n^n (n+1)!} \\
&= \lim_{n\to\infty} \frac{(n+1)^n}{n^n} \\
&= \lim_{n\to\infty} \left(\frac{n+1}{n}\right)^n \\
&= \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n \\
&= e > 1
\end{aligned}$$

$\frac{(n+1)^{n+1}}{(n+1)!} \cdot \frac{n^n}{n^n}$

$\left(\frac{n+1}{n}\right)^n = \left(1+\frac{1}{n}\right)^n$

$\to e > 1.$

diverges.

The Ratio Test shows that $\sum_{n=1}^{\infty} \frac{n^n}{n!}$ diverges.

**Question:** Does the series $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ converge or diverge?

# Root Test

**Note:** The following is a related to the Ratio Test with a similar proof.

## Theorem 1.5.4 (Root Test)

*Given a series* $\sum_{n=0}^{\infty} a_n$, *assume that*

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$$

*where* $0 \le L \in \mathbb{R}$ *or* $L = \infty$.

1. *If* $0 \le L < 1$, *then* $\sum_{n=0}^{\infty} a_n$ *converges absolutely.*
2. *If* $L > 1$, *then* $\sum_{n=0}^{\infty} a_n$ *diverges.*
3. *If* $L = 1$, *then no conclusion is possible.*

$\lim_{n\to\infty} \frac{1}{n} = 1$ → diverges

$\sum_{n=1}^{\infty} \frac{1}{n}$

no conclusion ←

$\sum_{n=1}^{\infty} \frac{1}{n^2}$

$\lim_{n\to\infty} \frac{1}{n^2} = 1$ → Converges

## Example 1.5.5

Determine if the series $\sum_{n=1}^{\infty}\left(\frac{3n^2+4}{4n^2+5n}\right)^n$ converges or diverges.

**Solution:** Let $a_n = \left(\frac{3n^2+4}{4n^2+5n}\right)^n$. Then

$$\lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\sqrt[n]{\left(\frac{3n^2+4}{4n^2+5n}\right)^n}$$

$$= \lim_{n\to\infty}\frac{3n^2+4}{4n^2+5n} \to \frac{3+\frac{4}{n^2}}{4+\frac{5}{n}}$$

$$= \frac{3}{4}$$

Since $\lim_{n\to\infty}\sqrt[n]{|a_n|} = \frac{3}{4} < 1$ we can use the Root Test to show that

$$\sum_{n=1}^{\infty}\left(\frac{3n^2+4}{4n^2+5n}\right)^n$$

converges.

## Example 1.5.6

Determine if the series $\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^{n^2}$ converges or diverges.

**Solution:** Let $a_n = \left(1+\frac{1}{n}\right)^{n^2}$. Then

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{\left(1+\frac{1}{n}\right)^{n^2}}$$

$$= \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^{n}$$

$$= e$$

Since $\lim_{n\to\infty} \sqrt[n]{|a_n|} = e > 1$ we can use the Root Test to show that

$$\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^{n^2}$$

diverges.

## Example 1.5.6

Determine if the series $\sum_{n=1}^{\infty}\left(1+\frac{1}{n}\right)^{n^2}$ converges or diverges.

**Solution:** Let $a_n=\left(1+\frac{1}{n}\right)^{n^2}$. Then

$$\begin{aligned}\lim_{n\to\infty}\sqrt[n]{|a_n|} &= \lim_{n\to\infty}\sqrt[n]{\left(1+\frac{1}{n}\right)^{n^2}}\\ &= \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n}\\ &= e\end{aligned}$$

Since $\lim_{n\to\infty}\sqrt[n]{|a_n|}=e>1$ we can use the Root Test to show that

$$\sum_{n=1}^{\infty}\left(1+\frac{1}{n}\right)^{n^2}$$

diverges.

## Example 1.5.6

Determine if the series $\sum_{n=1}^{\infty}\left(1+\frac{1}{n}\right)^{n^2}$ converges or diverges.

**Solution:** Let $a_n=\left(1+\frac{1}{n}\right)^{n^2}$. Then

$$\begin{aligned}\lim_{n\to\infty}\sqrt[n]{|a_n|} &= \lim_{n\to\infty}\sqrt[n]{\left(1+\frac{1}{n}\right)^{n^2}}\\ &= \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n}\\ &= e\end{aligned}$$

Since $\lim_{n\to\infty}\sqrt[n]{|a_n|}=e>1$ we can use the Root Test to show that

$$\sum_{n=1}^{\infty}\left(1+\frac{1}{n}\right)^{n^2}$$

diverges.